

LESSON 3-6 • SESSION 1 • PRACTICE • FORM A

Customary Unit Conversions

Learning target – I can convert between customary units by multiplying or dividing by the conversion rate. • 6.RP.A.3d

REMEMBER

- Converting changes the **unit**, not the amount. Write the unit fact first.
- Big unit to small unit: **multiply** ($5 \text{ ft} \times 12 = 60 \text{ in}$).
Small unit to big unit: **divide** ($12 \text{ qt} \div 4 = 3 \text{ gal}$).
- Facts: $1 \text{ ft} = 12 \text{ in}$ • $1 \text{ yd} = 3 \text{ ft}$ • $1 \text{ lb} = 16 \text{ oz}$ • $1 \text{ gal} = 4 \text{ qt}$ • $1 \text{ qt} = 2 \text{ pt}$ • $1 \text{ pt} = 2 \text{ cups}$

WORKED EXAMPLE

A pitcher holds 2 quarts. How many fluid ounces? ($1 \text{ qt} = 4 \text{ cups}$, $1 \text{ cup} = 8 \text{ fl oz}$)**Step 1** Quarts to cups: $2 \times 4 = 8 \text{ cups}$.**Step 2** Cups to fl oz: $8 \times 8 = 64 \text{ fl oz}$.**Step 3** Check: $64 \div 8 \div 4 = 2 \text{ quarts}$.**Answer 2 quarts = 64 fluid ounces****WORDS** customary units (*unidades usuales*) • convert (*convertir*) • conversion factor (*factor de conversión*)

PRACTICE

show your work • label units

1. Convert each measure.

a. $4 \text{ ft} = \underline{\hspace{1cm}} \text{ in}$

b. $5 \text{ yd} = \underline{\hspace{1cm}} \text{ ft}$

c. $4.5 \text{ lb} = \underline{\hspace{1cm}} \text{ oz}$

d. $20 \text{ qt} = \underline{\hspace{1cm}} \text{ gal}$

e. $56 \text{ oz} = \underline{\hspace{1cm}} \text{ lb}$

f. $3 \text{ gal} = \underline{\hspace{1cm}} \text{ qt}$

- 2.
- Roller Coaster.**
- Riders must be at least 48 inches tall. Caleb is 4 feet tall. Complete the table. Is Caleb tall enough?

Hint Each foot adds 12 inches.

Feet	1	2	3	4
Inches	12	24	36	

Tall enough? _____

- 3.
- Party Punch.**
- A punch bowl holds 2 gallons. (
- $1 \text{ gal} = 4 \text{ qt}$
- ,
- $1 \text{ qt} = 4 \text{ cups}$
-)

a. Quarts: _____ qt

b. Cups: _____ cups

Show your work.

- 4.
- Gym Rope.**
- A climbing rope is 8 yards long. (
- $1 \text{ yd} = 3 \text{ ft}$
- ,
- $1 \text{ ft} = 12 \text{ in}$
-)

a. Feet: _____ ft

b. Inches: _____ in

Show your work.

- 5.
- Road Race.**
- A road race is 2.5 miles long. How many feet is the race? (
- $1 \text{ mile} = 5,280 \text{ feet}$
-)

Hint Miles to feet is big to small: multiply by 5,280.

Show your work.

Answer _____ feet

- 6. Water Cooler.** A cooler holds 9 gallons of water. How many 1-pint cups can it fill? (1 gal = 4 qt, 1 qt = 2 pt)

Hint Change gallons to quarts first, then quarts to pints.

Show your work.

Answer _____ cups

- 7. Milk Cartons.** The cafeteria has 24 quarts of milk. How many gallons is that? (4 qt = 1 gal)

Hint Quarts to gallons is small to big: divide.

Show your work.

Answer _____ gallons

- 8. Garden Fence.** A garden needs 24 feet of fencing. Fencing is sold by the yard at \$9 per yard. (1 yd = 3 ft)

a. Yards needed: _____ yd

b. Total cost: _____

c. Would buying by the foot at \$3.25 per foot cost more? How much? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Sam's Work.** Sam converted 60 inches to feet by multiplying: $60 \times 12 = 720$ feet. Explain why 720 feet is impossible. Then find the correct answer.

TEST PRACTICE

choose the best answer

- 10.** 5 feet equals how many inches?

A 17

B 50

C 60

D 72

- 11.** 48 ounces equals how many pounds?

A 768

B 32

C 4

D 3

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

13,200 H	6 S	32 N	18 X	2,112 Q	48 I	72 E	96 Z	288 C
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#1a	#3b	#4b	#5	#6	#7
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____